

## Experiment-3

### Python code

```
In [2]: import cv2
import numpy as np
import matplotlib.pyplot as plt
import time

# STEP 1: Load the image
image = cv2.imread("image.jpg.png")

# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert BGR image to RGB for Matplotlib
original = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# STEP 2: Apply Gaussian Filter
gaussian = cv2.GaussianBlur(image, (5, 5), 0)

# STEP 3: Apply Median Filter
median = cv2.medianBlur(image, 5)

# STEP 4: Apply Average (Mean) Filter
average_kernel = np.ones((5, 5), np.float32) / 25
average = cv2.filter2D(image, -1, average_kernel)

# STEP 5: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# STEP 6: Apply Laplacian Filter
laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

# STEP 7: Apply Sobel Edge Detection
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

# Combine Sobel X and Y
sobel_combined = cv2.addWeighted(
    sobel_x, 0.5,
    sobel_y, 0.5,
    0
)

# Convert filtered images to RGB
gaussian_rgb = cv2.cvtColor(gaussian, cv2.COLOR_BGR2RGB)
median_rgb = cv2.cvtColor(median, cv2.COLOR_BGR2RGB)
```

```
average_rgb = cv2.cvtColor(average, cv2.COLOR_BGR2RGB)

# STEP 8: Display all images
plt.figure(figsize=(15, 10))

plt.subplot(2, 4, 1)
plt.imshow(original)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 4, 2)
plt.imshow(gaussian_rgb)
plt.title("Gaussian Filter")
plt.axis("off")

plt.subplot(2, 4, 3)
plt.imshow(median_rgb)
plt.title("Median Filter")
plt.axis("off")

plt.subplot(2, 4, 4)
plt.imshow(average_rgb)
plt.title("Average Filter")
plt.axis("off")

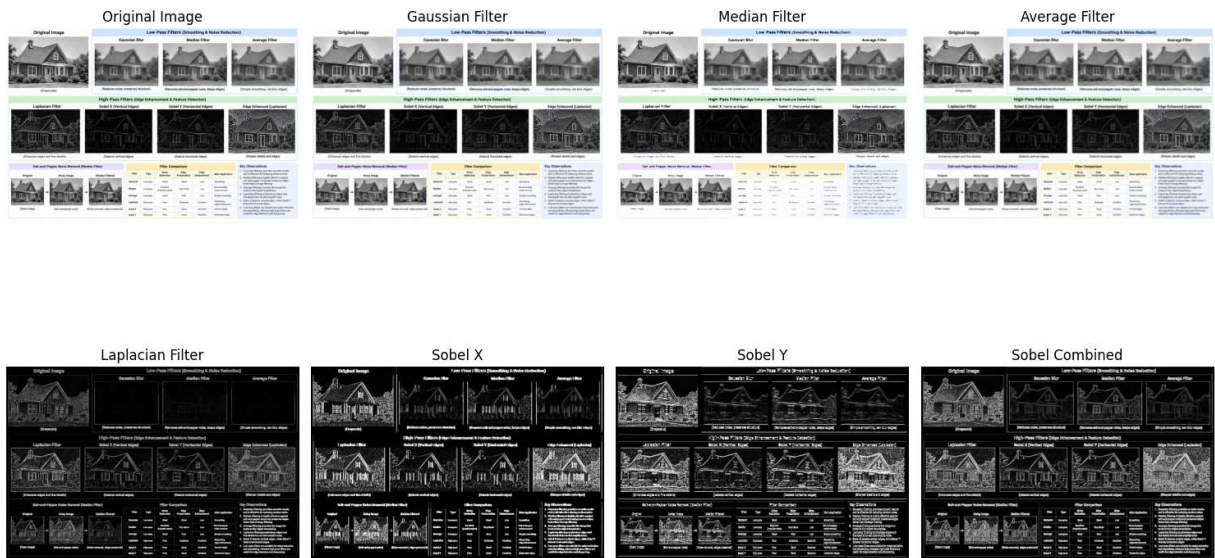
plt.subplot(2, 4, 5)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian Filter")
plt.axis("off")

plt.subplot(2, 4, 6)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X")
plt.axis("off")

plt.subplot(2, 4, 7)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y")
plt.axis("off")

plt.subplot(2, 4, 8)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Sobel Combined")
plt.axis("off")

plt.tight_layout()
plt.show()
```



```
In [4]: #step 9 Compare computational performance
start = time.time()
cv2.blur(image, (5, 5))
average_time = time.time() - start

start = time.time()
cv2.GaussianBlur(image, (5, 5), 0)
gaussian_time = time.time() - start

start = time.time()
cv2.medianBlur(image, 5)
median_time = time.time() - start

start = time.time()
cv2.Laplacian(gray, cv2.CV_64F)
laplacian_time = time.time() - start

start = time.time()
cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_time = time.time() - start

print("\n--- Computational Performance ---")
print("Average Filter Time:", average_time, "seconds")
print("Gaussian Filter Time:", gaussian_time, "seconds")
print("Median Filter Time:", median_time, "seconds")
print("Laplacian Filter Time:", laplacian_time, "seconds")
print("Sobel Filter Time:", sobel_time, "seconds")

#step 10 Display observations and interpretations

print("\n--- Observations ---")

print("1. Average Filter:")
print("    Reduces noise but may blur edges.")

print("\n2. Gaussian Filter:")
print("    Provides smooth results and preserves structures better.")
```

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print("\n3. Median Filter:")
print("    Best for removing salt-and-pepper noise.")

print("\n4. Laplacian Filter:")
print("    Detects edges and enhances fine details.")

print("\n5. Sobel Filter:")
print("    Detects horizontal and vertical edges.")

print("\n--- Conclusion ---")
print("Low-pass filters are useful for image smoothing and noise reduction.")
print("High-pass filters are useful for edge detection and image sharpening.")
print("The choice of filter depends on the application and image type.")

```

#### --- Computational Performance ---

Average Filter Time: 0.013513326644897461 seconds  
 Gaussian Filter Time: 0.0019910335540771484 seconds  
 Median Filter Time: 0.013001203536987305 seconds  
 Laplacian Filter Time: 0.01201319694519043 seconds  
 Sobel Filter Time: 0.01212000846862793 seconds

#### --- Observations ---

1. Average Filter:  
Reduces noise but may blur edges.
2. Gaussian Filter:  
Provides smooth results and preserves structures better.
3. Median Filter:  
Best for removing salt-and-pepper noise.
4. Laplacian Filter:  
Detects edges and enhances fine details.
5. Sobel Filter:  
Detects horizontal and vertical edges.

#### --- Conclusion ---

Low-pass filters are useful for image smoothing and noise reduction.  
 High-pass filters are useful for edge detection and image sharpening.  
 The choice of filter depends on the application and image type.

## Questions and Answers

### 1. What is spatial filtering? How is it used in digital image processing?

Spatial filtering is a technique in which the value of a pixel is modified based on its neighboring pixels using a filter or kernel. The kernel is moved across the image to perform operations such as smoothing, sharpening, noise removal, and edge detection.

It is commonly used as a preprocessing step to improve image quality and prepare images for further computer vision tasks.

### 2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.

| Low-Pass Filter                          | High-Pass Filter                          |
|------------------------------------------|-------------------------------------------|
| Allows low-frequency components to pass. | Allows high-frequency components to pass. |
| Reduces noise and smooths the image.     | Enhances edges and fine details.          |
| Produces a blurred or smoother image.    | Produces a sharper image.                 |
| Example: Average and Gaussian filters.   | Example: Sobel and Laplacian filters.     |

### 3. Compare Average Filter, Gaussian Filter, and Median Filter based on their working principles and applications.

| Filter                 | Working Principle                                                   | Applications                                           |
|------------------------|---------------------------------------------------------------------|--------------------------------------------------------|
| <b>Average Filter</b>  | Replaces each pixel with the average of its neighboring pixels.     | Image smoothing and noise reduction.                   |
| <b>Gaussian Filter</b> | Uses a weighted average where nearby pixels have higher importance. | Smoothing and reducing Gaussian noise.                 |
| <b>Median Filter</b>   | Replaces each pixel with the median value of its neighborhood.      | Removing salt-and-pepper noise while preserving edges. |

The Median Filter generally preserves edges better than the Average Filter during noise removal.

### 4. Why is the Median Filter particularly effective for removing salt-and-pepper noise?

The Median Filter replaces a pixel with the median value of its surrounding pixels. Salt-and-pepper noise contains extreme values such as very bright or very dark pixels.

Since the median is less affected by these extreme values than the average, the noise can be removed effectively while preserving important image edges.

### 5. Explain the role of convolution kernels in spatial filtering.

A convolution kernel is a small matrix of numerical values used to process an image. The kernel moves across the image and performs mathematical operations with neighboring pixel values.

Different kernels produce different effects. For example, smoothing kernels reduce noise, while edge-detection kernels such as Sobel highlight changes in intensity and detect edges.

### 6. What is the purpose of the Sobel and Laplacian operators in edge detection?

The **Sobel operator** detects edges by calculating intensity changes in horizontal and vertical directions. It can also provide information about the direction of edges.

The **Laplacian operator** uses the second derivative of image intensity to detect rapid changes in intensity. It can detect edges in different directions but is more sensitive to noise.

## 7. Why are filtering operations considered an essential preprocessing step in computer vision?

Filtering improves the quality of images before applying computer vision algorithms. It can remove noise, reduce unwanted details, smooth images, sharpen important features, and enhance edges.

This helps subsequent tasks such as segmentation, object detection, feature extraction, and image recognition produce more reliable results.

## 8. Discuss the trade-off between image smoothing and edge preservation during filtering.

Image smoothing reduces noise and unwanted variations, but excessive smoothing can blur important edges and fine details.

On the other hand, strong edge preservation keeps object boundaries clear but may also retain unwanted noise. Therefore, the appropriate filter and kernel size should be selected according to the application.

## 9. Mention four real-world applications where spatial filtering techniques are widely used.

Four real-world applications are:

1. **Medical Imaging:** Removing noise and enhancing edges in X-ray, CT, and MRI images.
2. **Satellite Imaging:** Improving image quality and detecting geographical features.
3. **Security and Surveillance:** Enhancing video frames and detecting object boundaries.
4. **Photography:** Reducing noise, smoothing images, and sharpening details.

## 10. Compare spatial domain filtering with frequency domain filtering in terms of implementation and practical applications.

| Spatial Domain Filtering                                            | Frequency Domain Filtering                                               |
|---------------------------------------------------------------------|--------------------------------------------------------------------------|
| Operates directly on image pixels.                                  | Operates on frequency components of an image.                            |
| Uses convolution kernels or masks.                                  | Uses transforms such as Fourier Transform.                               |
| Usually easier to implement.                                        | Generally more complex to implement.                                     |
| Suitable for local operations such as smoothing and edge detection. | Useful for controlling specific frequency components.                    |
| Examples: Average, Gaussian, Median, Sobel.                         | Examples: Frequency-domain Low-Pass and High-Pass Filters.               |
| Commonly used for noise removal, sharpening, and edge detection.    | Commonly used for periodic noise removal and advanced image enhancement. |